Organize a Fabric Lakehouse using medallion architecture design

Core Architecture Layers

The medallion architecture typically consists of three progressively refined layers:

- **Bronze (Raw):** The initial landing zone for all structured, semi-structured, or unstructured data. Data is stored in its original format without any changes.
- **Silver (Validated):** The layer where data is cleansed, deduplicated, and validated. It serves as a central repository where data is stored in a consistent format for multiple teams.
- **Gold (Enriched):** The presentation layer where data is aggregated and modeled (often in a star schema) to meet specific business and analytics needs.

Implementation in Microsoft Fabric

- **Tools for Data Movement:**
  - **Pipelines:** Used for data orchestration and moving data between layers.
  - **Dataflows (Gen2):** Best for smaller semantic models and simple transformations.
  - **Notebooks:** Ideal for large-scale data and complex transformations; they allow saving data as managed Delta tables.
- **Data Storage:** Fabric lakehouses are built on the **Delta Lake** format, which supports ACID transactions.
- **Flexibility:** Users can add additional layers (e.g., "Platinum") or create multiple Gold layers tailored to specific departments like Finance or Sales.

Querying and Reporting

- **SQL Analytics Endpoint:** Provides a read-only T-SQL experience to explore and query data in Delta tables at any layer.
- **Direct Lake Mode:** Enables Power BI to access data directly from the Lakehouse Delta tables without needing to import or refresh the data.
- **Semantic Models:** Automatically created with a lakehouse, these allow data to be viewed using business-friendly terminology.

Security and Governance

- **Permissions:** Security is managed through workspace-level permissions or item-level permissions for more granular control.
- **Access Control:** Best practices recommend restricting the Bronze and Gold layers to read-only access for most users to maintain data integrity.
- **CI/CD:** Fabric integrates with Git to allow data teams to version work, collaborate via branches, and automate deployments. This is especially critical for the Gold layer to ensure reliable data for consumption.

Questions

1. Which of the following sets of layers are typically associated with the Medallion Architecture for data management?
A. Raw, Polished, Refined
B. Bronze, Silver, Gold
C. Initial, Intermediate, Final
D. Primary, Secondary, Tertiary

2. Which tool is best suited for data transformation in Fabric when dealing with large-scale data that will continue to grow?
A. Dataflows (Gen2)
B. Pipelines
C. Notebooks
D. SQL analytics endpoint

3. What is the benefit of storing different layers of your lakehouse in separate workspaces?
A. It can enhance security, manage capacity use, and optimize cost-effectiveness.
B. It makes it easier to share data with colleagues.
C. There's no benefit of storing different layers of your lakehouse in separate workspaces.
D. It automatically models the data into a star schema.

4. Which layer of the medallion architecture acts as the landing zone where data is stored in its original format?
A. Silver
B. Platinum
C. Gold
D. Bronze

5. What is the primary purpose of the Silver layer in a lakehouse environment?
A. To store unstructured data for the first time
B. To validate, refine, and cleanse data through deduplication and null removal
C. To provide a read-only presentation layer for executive reporting
D. To serve as a temporary staging area that is deleted daily

6. Which technology natively supports ACID transactions in a Fabric lakehouse?
A. CSV files
B. Delta Lake
C. JSON format
D. Parquet (non-Delta)

7. When data reaches the Gold layer, it is typically modeled into which format for optimized reporting?
A. Flat file
B. Snowflake schema
C. Star schema
D. Unstructured document store

8. Which tool is specifically described as the primary choice for data orchestration in Fabric?
A. Dataflows
B. Pipelines
C. Notebooks
D. Power BI

9. Which Power BI feature allows data analysts to access data directly from Delta tables without importing or refreshing it?
A. Import mode
B. DirectQuery
C. Direct Lake mode
D. Live Connection

10. What is a "Platinum" layer in the context of customizing the medallion architecture?
A. A mandatory fourth layer for all lakehouses
B. An optional layer for data further refined for a specific use case
C. A backup layer for disaster recovery
D. The layer used specifically for storing raw JSON files

11. What is the recommended access level for most users when interacting with the Bronze and Gold layers?
A. Read/Write
B. Owner
C. Read-only
D. Contributor

12. In the medallion architecture, "multi-hop" refers to what process?
A. Moving data between different cloud providers
B. The movement of data between layers as it increases in quality
C. Connecting multiple Power BI reports to one source
D. Switching between Spark and SQL engines

13. Which component of the Fabric Lakehouse allows you to use T-SQL to query data in read-only mode?
A. Data Factory
B. SQL analytics endpoint
C. KQL Database
D. Semantic Link

Answers

1. B
2. C
3. A
4. D
5. B
6. B
7. C
8. B
9. C
10. B
11. C
12. B
13. B